

Preliminary Patentability Assessment

Topological Cross-Domain Corpus

Can the work be patented - and can it be patented as a whole?

PREPARED FOR

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ASSESSMENT DATE

14 August 2026

JURISDICTIONAL FOCUS

Netherlands and European Patent Convention (EPC)

SOURCE REVIEWED

Topological Cross-Domain Synthesis of the Spherical Substrate Corpus, version 2.0, 63 pages

STATUS

Confidential - preliminary, non-exhaustive assessment

PRIVACY NOTE

Personal identifiers supplied with the request are intentionally omitted from this report because they do not affect novelty, inventorship, ownership, or patentability. Keep identity documents and tax or registration numbers out of patent drafts unless a filing form specifically requires them.

LEGAL STATUS

This report is not a legal opinion, a complete novelty search, a freedom-to-operate analysis, or a substitute for advice from a Dutch or European patent attorney. It is an evidence-based triage of what is worth preserving, testing, and presenting to counsel.

SECTION 0

Executive decision

The source contains several possible patent projects, but not one credible all-encompassing patent.

BOTTOM LINE

Yes: parts of the work may support patent applications if they are narrowed to concrete technical implementations and backed by enabling detail and measured technical effects. No: the corpus as a whole is unlikely to form a strong single patent application. It spans several technical fields, contains abstract and expressly rejected propositions, and currently lacks the implementation detail and validation needed for its broadest claims.

Family	Proposed subject	Preliminary potential	Readiness	Recommended treatment
A	Compatibility-gated certified event-solving engine	Medium	2/5	Preserve now; prototype and benchmark before filing broad claims
B	Hollowlens-0 bounded compatibility event detector/controller	Medium to medium-high after engineering	1/5	Strongest future hardware family; urgent engineering disclosure needed
C	Novelty-proportional state reconstruction and lineage storage	Low to medium	2/5	Keep as dependent claims or separate narrow storage/computer-architecture case
D	KC log-polar, SDF-edge-locked 1-bit projection/prepress	Low	1-2/5	Only pursue if an exact algorithm and comparative image/print data emerge
E	Double vacuum, hourglass, parity, Klein/Mobius/golden-ratio motifs alone	Very low	1/5	Use only as implementation vocabulary; do not lead with these abstractions
Whole	Entire cross-domain corpus in one application	Very low	1/5	Do not file as one totalizing invention; use a portfolio strategy

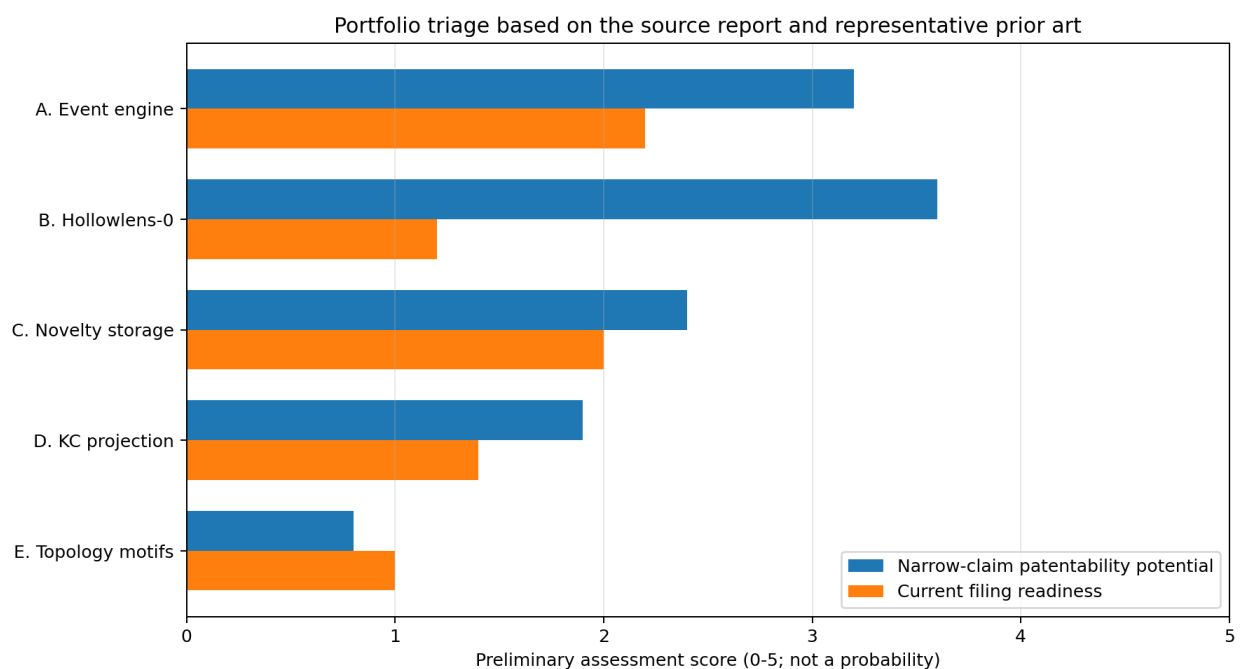


Figure 1. Portfolio triage. Scores are qualitative and are not legal probabilities.

Recommended decisions now

1. Stop further public distribution until a patent attorney has reviewed the disclosure history. The Dutch patent authority states that an invention must not have been made public anywhere before filing, including by the inventor, and specifically warns against entering sensitive invention details into AI systems [NL1].
2. Prepare one software disclosure around a technical event-processing engine, not around an ontology, a universe model, or a renderer replacement. The strongest common thread is support and compatibility filtering before certified event solving, followed by an auditable typed transition.
3. Prepare a separate hardware disclosure for Hollowlens-0 only after defining the physical stack, dimensions, materials, wavelength, coupling geometry, control loop, calibration sequence, error boundary, and measured crosstalk/false-event performance.
4. Treat memory/lineage and KC projection as secondary families or dependent claim sets unless experiments reveal a distinct technical advantage.
5. Do not claim zero memory, zero heat, zero latency, universal $O(1)$, biological control, physical untouchability, or topology as a substitute for physical equations. The source itself rejects or excludes these propositions.

What this assessment does not determine

- Whether any source PDF, web post, presentation, repository, sale, demonstration, or third-party disclosure already destroyed novelty.
- Whether Tom Klootwijk is the sole inventor of every proposed claim. Inventorship is claim-specific and depends on human contributions to the inventive concepts.
- Whether commercial products can be made or sold without infringing third-party patents. Patentability and freedom to operate are separate questions.
- Whether a competent authority would grant any particular claim. A full search, claim drafting, and prosecution strategy remain necessary.

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SECTION 1

Mandate, source, and method

A source-grounded patentability triage, not a replacement for prosecution counsel.

1.1 Question assessed

The mandate is to determine whether any part of the work described in the supplied report may be patentable, whether the work could be protected as a whole, which embodiments deserve priority, and what evidence is missing before a credible Dutch or European filing. The analysis is directed primarily to the Netherlands and the European Patent Convention because the requester supplied a Dutch context and identifier.

1.2 Source reviewed

The reviewed source is a 63-page report titled Topological Cross-Domain Synthesis of the Spherical Substrate Corpus, version 2.0, dated 14 August 2026. It says it reconstructs seven user-supplied PDFs and treats the mature center as a queryable state-and-event substrate. The report expressly states that it has no external performance validation and that it is not itself patent analysis.

Source note: Source report pp. 1-2 and 63. File SHA-256: dde0ba0961c46faed2f0ed970eee7a8ce5963f7e43f648de4f72ddb3a826f4c8.

1.3 Method

1. Extract the source-defined technical architecture and separate it from metaphors, rejected claims, and report-level information-science normalizations.
2. Group the material into potential invention families that could plausibly satisfy unity of invention.
3. Apply current EPC criteria for patentable subject matter, novelty, inventive step, sufficiency, industrial applicability, unity, and entitlement.
4. Compare each family against representative academic and patent prior art. This is a landscape review, not an exhaustive search.
5. Assess filing readiness: whether the current document contains enough detail to draft claims without impermissibly adding new matter later.
6. Recommend experiments, documentation, claim architecture, filing sequence, and non-patent alternatives.

1.4 Confidence and limitations

IMPORTANT

A patentability conclusion is claim-specific. A phrase such as “event calculus” is neither patentable nor unpatentable in the abstract. The legal question is whether a particular claim defines a new and non-obvious technical solution, and whether the application enables that solution over the claimed scope.

- No interview was conducted with the human contributors. Inventorship and ownership conclusions are therefore provisional.
- No non-public lab notebooks, code repository, hardware drawings, test data, or disclosure log were provided.
- Patent databases contain unpublished applications for approximately 18 months after filing; no search can eliminate that uncertainty.
- Representative references were selected to identify crowded areas and drafting risks, not to prove anticipation of every possible claim.

SECTION 2

Patentability framework in the Netherlands and Europe

The relevant filters are technical character, novelty, inventive step, sufficiency, industrial applicability, and unity.

2.1 Core EPC requirements

Requirement	Practical meaning for this corpus	Authority
Patentable invention / technical character	The claimed subject must be an invention in a field of technology. Mathematical methods, aesthetic creations, and computer programs “as such” are excluded, but a computer-implemented method can qualify when it makes a technical contribution.	EPC Art. 52; EPO Guidelines G-II 3.3 and 3.6
Novelty	The claim must not be disclosed in a single item of prior art. Everything made available to the public before filing can form part of the state of the art.	EPC Art. 54
Inventive step	The claim must not be obvious to the skilled person in view of the prior art. For mixed technical/non-technical claims, a non-obvious technical solution to a technical problem is required.	EPC Art. 56; COMVIK approach
Sufficiency	The application must teach the skilled person how to carry out the invention without an undue research programme across the claimed scope.	EPC Art. 83
Industrial applicability	The subject must be capable of being made or used in industry. Concrete software control, simulation, sensing, graphics processing, and hardware embodiments can satisfy this requirement.	EPC Art. 52(1) and 57
Unity	One application must concern one invention or a group linked by a single general inventive concept.	EPC Art. 82

2.2 Software, mathematics, and simulations

The source is strongest when read as computer-implemented event processing. That is not automatically excluded. The EPO recognizes that a mathematical method can contribute to technical character when it is applied in a technical field or adapted to a specific technical implementation [E7]. A computer program may produce a further technical effect when it controls a technical system or improves the computer’s internal operation, for example processor load balancing or memory allocation [E9].

The difficulty is inventive step. Merely putting an abstract ontology, data schema, mathematical model, or business rule on a computer does not make the non-technical content inventive. Under the COMVIK approach, only features contributing to a technical effect can support inventive step [E10]. A simulation is assessed in the same way: the technical character of the simulated subject alone is not decisive; the claimed method must produce a technical effect or be specifically adapted to a technical implementation [E8].

DRAFTING CONSEQUENCE

Do not lead with “a world model,” “a universal equation world,” “identity ontology,” or “topological substrate.” Lead with a technical process: which machine or physical system is processed, which data structures and numerical operations are used, where computational work is avoided, how correctness is certified, and what measurable resource or control effect results.

2.3 Public disclosure and the European no-grace-period problem

For a European filing, public disclosure before filing is normally fatal to novelty. EPC Article 55 provides only narrow six-month exceptions for evident abuse or certain official exhibitions; it is not a general inventor grace period [E6]. Octrooiencentrum Nederland likewise states that an invention must not have been made public anywhere, even by the inventor, and advises keeping it secret until filing [NL1].

2.4 Dutch filing route

A Dutch application can establish a priority date and requires a prior-art search. It is published and granted at 18 months and can last up to 20 years, but the current Dutch system can grant even when the search is negative; validity may later be challenged in court [NL3, NL7]. A Dutch first filing can be extended to a European or PCT route within the 12-month priority year [NL5].

2.5 Inventor and applicant

The right to a European patent belongs to the inventor or successor in title, subject to employee-invention law where applicable [E11]. Inventorship is determined by contribution to the inventive concept of the claims, not by ownership of a document, registration numbers, date of birth, project management, or payment. The EPO has held that a machine is not an inventor under the EPC [E12]. Human contributors must therefore be identified claim by claim, while AI assistance should be documented as a tool rather than named as inventor.

SECTION 3

Technical reading of the source document

The source itself supplies a disciplined distinction between a viable core and speculative totalizing claims.

3.1 The mature core

The report’s strongest formulation is a “spherical-support event calculus.” A finite grammar and exogenous log define or reconstruct state; local radial-angular supports restrict relevance; a compatibility predicate determines whether sectors may couple; explicit relation surfaces define events; a root solver identifies the earliest admitted event; and typed transitions update state while preserving invariants and lineage. Images, meshes, raster output, vectors, print, and photonic readout are downstream views rather than the authoritative state.

Source note: Source report pp. 4, 9, 13-17, and 43. The report’s event equation requires relation satisfaction, support admission, and compatibility.

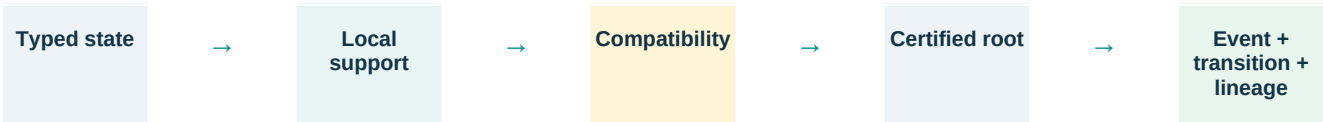


Figure 2. Patent-relevant core pipeline extracted from the source report.

3.2 Source-defined state and event record

- State is a product including position or projective geometry, continuous time, phase, sheet/orientation, lineage address, branch, and a narrow parity/route bit.
- An event is not a mere geometric hit. It exists when a relation or guard reaches zero, the state lies in a declared support, and a versioned compatibility predicate admits coupling.
- The transition is typed and is intended to preserve declared invariants and lineage.
- The event record includes relation and transition identifiers, root time, support and compatibility results, predicate version, confidence, pre/post hashes, lineage parent, branch creation, parity change, and an invariant report.

3.3 Four separable technical domains

Domain	Source components	Patent family
Software event engine	State, relation, support, compatibility, certified root, transition, event certificate	Candidate A
Storage/provenance architecture	Seed and grammar reconstruction, exogenous log, branches, lineage hashes, materialized views	Candidate C
Graphics and prepress adapter	Log-polar addressing, SDF/CSG, edge accumulation, 1-bit PDM/dither, CMYK and dot-gain compensation	Candidate D
Optofluidic hardware	Liquid coupler, interferometric waveguide, balanced detection, calibration and event-control sidecar	Candidate B

3.4 What the source itself demotes or rejects

The source is unusually helpful because it does not insist that every motif is literal. It demotes Klein-bottle and Mobius language to explicit gluing or orientation maps, treats the hourglass as a routing visualization, confines one bit to a narrow route/parity role, and translates “double vacuum” into co-location without coupling. It rejects universal O(1), zero memory, zero latency, zero heat, literal topological immunity, and direct biological or medical control. These boundaries should be retained in any patent strategy rather than reversed.

SECTION 4

Overall portfolio answer

A portfolio is plausible; one totalizing patent is not.

4.1 Can any of the work be patented?

Potentially yes. The best route is not to patent the corpus's vocabulary or worldview. It is to identify one or more concrete technical implementations whose distinguishing features produce a measurable technical effect. The strongest candidates are: (A) a compatibility-gated certified event processor for a physical-system simulation or control task; and (B) a specifically engineered optofluidic event detector with a calibrated control sidecar.

Candidate C may support narrow computer-architecture or storage claims if it demonstrably reduces memory, input/output, or replay cost. Candidate D may support a narrow imaging or preprocess algorithm only if the exact mask, edge-locking, compensation, and objective output advantage are disclosed. The topological motifs are generally better used as terminology in the description or as dependent implementation features.

4.2 Can it be patented as a whole?

Not credibly in its current form. A single application must relate to one invention or a linked group sharing one general inventive concept. The source crosses software simulation, database provenance, graphics/prepress, optics, fluid mechanics, photonic hardware, identity semantics, and explicitly excluded biological or personal narratives. The common vocabulary is not enough: unity normally turns on shared special technical features that define a contribution over the prior art.

An umbrella application could possibly contain software and hardware embodiments if every independent claim is centered on the same technical feature - for example, a support-and-compatibility-gated event certification protocol. Even then, the graphics and storage families may be found non-unitary and require divisionals. A portfolio planned from the outset is more robust.

4.3 Recommended portfolio architecture

Portfolio element	Core subject	Claim categories	Trigger
Priority filing A	Technical event processor	Software method/system/program/medium; optional physical-system control embodiment	File when exact algorithm, benchmarks, and disclosure audit are ready
Priority filing B	Hollowlens-0 event detector/controller	Apparatus and operating method; possibly calibration method	File after physical design is enabling, or earlier only if disclosure risk and the specification can support claims
Optional filing C	Reconstruction and lineage storage	Computer-architecture/storage method	Separate only if benchmarks show a distinct technical storage/IO effect
Optional filing D	Projection/prepress algorithm	Image-processing method and output device/control claims	Separate only if exact algorithm and comparative quality metrics exist
Trade secret layer	Solver heuristics, compatibility policy tuning, calibration coefficients, benchmark datasets	Confidential know-how	Keep secret where reverse engineering is difficult and patent disclosure would not add strategic value

SECTION 5

Candidate A - compatibility-gated certified event engine

The best near-term software family, provided it is framed as a technical processor and not an abstract ontology.

5.1 Proposed inventive concept

A computer-implemented event-processing method for a model of a physical or technical system, in which candidate relation surfaces are first restricted by an analytic local support and a versioned compatibility predicate; only admitted candidates undergo certified root solving to determine the earliest event; the system then emits an auditable event certificate and applies a typed transition with invariant and lineage checks.

Source note: Source report pp. 16-17, 26-27, 34-38, and 43. The report's worked example solves the event as a root, rejects incompatible co-located sectors, toggles parity, creates a branch, and records before/after hashes.

5.2 Why it has potential

- It can be tied to a technical use, such as continuous collision monitoring, robotic control, process control, network timing, photonic channel control, or simulation of a specific physical device.
- The proposed ordering may reduce expensive numerical work: support and compatibility can prune candidates before certified root refinement.
- The event certificate can improve deterministic replay, fault diagnosis, and safety audit by recording solver certainty, predicate version, state hashes, and invariant outcomes.
- A technical effect may also arise from computer-internal improvements if the claimed data structures and scheduling reduce processor, memory, or input/output load.

5.3 Crowded prior-art field

Broad claims are vulnerable. Hybrid automata have long combined continuous state with discrete modes, guards, and transitions [PA1, PA2]. Quantized State Systems use a discrete-event approach for continuous-time systems [PA3]. Continuous collision detection computes event or contact times rather than relying solely on frame stepping [PA4]. Patents also describe hybrid automata, continuous/discrete simulation, triggered state changes, and formal translation of hybrid models [PP1-PP3].

Accordingly, the source's broad sequence "state + guard + earliest root + transition" is unlikely to be novel. The possible inventive residue is narrower: the precise support/compatibility prefilter, certification contract, event-record schema, deterministic ordering, and technical resource effect in a defined application.

5.4 Patentability assessment

Issue	Assessment
Technical character	Medium to high when tied to a physical-system simulation/control task or a demonstrated computer-internal resource improvement; low if claimed as a general ontology.
Novelty	Low for broad hybrid state/guard/transition concepts. Medium only for a specific combination and ordering not disclosed in a single reference.
Inventive step	Low to medium. Prefiltering before an expensive solver can be an obvious optimization unless the claimed predicate/support structure and measured effect are non-obvious.
Sufficiency	Medium for the restricted Equation World Zero example, but low for broad relation families. The application must teach root certification, degeneracies, event ordering, branch handling, and performance boundaries.
Unity	Good as one software family if all claims share the same event certification pipeline.

Issue	Assessment
Current readiness	2/5. The source has architecture and a restricted prototype description, but no supplied code, benchmark data, or detailed solver specification in the reviewed PDF.

5.5 What must be added before filing

1. Choose one technical application and define the physical quantities represented by the state, relations, supports, and transitions.
2. Specify the relation family and certified root method, including interval bounds, multiplicity/tangency handling, no-root certificates, tolerances, and tie-breaking.
3. Specify the support representation and exactly when support can be evaluated before root solving without missing events.
4. Specify the compatibility predicate inputs, reason codes, versioning, caching, update rules, and causal relationship to the technical system.
5. Define the event-certificate data structure and explain how each field contributes to fault tolerance, deterministic replay, or control safety.
6. Benchmark against at least two baselines: frame/time stepping and a conventional event queue or continuous-collision method, at equal correctness and uncertainty.
7. Report candidate counts before support, after support, and after compatibility; root-solver calls; CPU/GPU time; memory; miss/false-event rates; and tail latency.
8. Include multiple embodiments and failure cases so the disclosure is not merely an invitation to conduct a research programme.

5.6 Drafting position

GO, WITH CONDITIONS

Preserve this family now. Do not file a broad “equation world” claim. File only after the technical field, exact pipeline, and comparative effect are documented, unless an imminent disclosure forces an earlier priority filing that is still enabling.

SECTION 6

Candidate B - Hollowlens-0 optofluidic event detector

Potentially the strongest future technical family, but currently the least enabled.

6.1 Proposed inventive concept

An optofluidic apparatus comprising a defined optical input, a tunable liquid coupler, an interferometric waveguide structure, balanced detectors, and a controller that declares a bounded compatibility event only when a spatial or angular support condition, channel-compatibility condition, measured guard crossing, and confidence threshold are all met; the controller also performs calibration, drift monitoring, parity/routing, and event lineage logging.

Source note: Source report pp. 39-40 and 45. The source frames Hollowlens-0 as an optional bounded bridge, not proof of the software ontology or of a system advantage.

6.2 Why it has potential

- Hardware and control claims naturally have technical character.
- A specific physical arrangement, actuation mechanism, calibration sequence, or event-detection protocol can be patentable even when its individual optical components are known.
- The source's four-condition event declaration may provide a coherent control protocol if it demonstrably reduces false events, crosstalk, calibration burden, or energy per verified event.
- Apparatus, operating method, calibration method, and control-software claims can be layered in one family if they share the same special technical feature.

6.3 Crowded prior-art field

Generic optofluidic tuning and waveguide detection are crowded. Representative patent literature includes reconfigurable liquid-infiltrated optical waveguides [PH1], optofluidic multi-mode-interference analyte detection [PH2], programmable photonic processing [PH3], self-configuring or field-programmable interferometers [PH4], and calibration/monitoring/control of integrated photonic systems [PH5]. A claim that merely combines a liquid coupler, interferometer, detector, and digital controller would therefore be weak.

6.4 Patentability assessment

Issue	Assessment
Technical character	High.
Novelty	Unknown. A unique geometry, materials stack, actuation path, mode-selection arrangement, or calibration sequence is not yet disclosed in sufficient detail.
Inventive step	Potentially medium if the four-condition event protocol is causally tied to a measured device improvement and is not a routine aggregation of known thresholds and controls.
Sufficiency	Currently low. The report does not provide fabrication-ready dimensions, materials, wavelength ranges, fluid properties, electrode/actuator design, transfer functions, control law, or measured data.
Unity	Good if apparatus and controller claims share a specific physical compatibility-event mechanism.
Current readiness	1/5. It is a research programme and minimum concept, not yet an enabling patent specification.

6.5 Required engineering disclosure

1. Optical wavelength(s), source type, launch geometry, polarization, numerical aperture, and input scan or addressing method.
2. Waveguide material stack, dimensions, refractive indices, coupling length, interferometer topology, detector type, and balanced readout circuit.
3. Liquid composition, containment, meniscus or interface geometry, actuation method, control range, response time, hysteresis, temperature dependence, and packaging.
4. The exact physical meaning of support, compatibility, guard, confidence, and parity in the device.
5. Calibration and drift-correction sequence, reference channels, failure detection, fallback mode, and stored calibration state.
6. Comparative data at equal error: crosstalk, false/missed events, event rate, energy per verified event, settling time, thermal drift, and calibration overhead.
7. At least one working embodiment plus variants that support the intended claim breadth.

6.6 Drafting position

DEVELOP FIRST; FILE WHEN ENABLING

This may become the most defensible patent family because it is concrete hardware. Today, however, the source mainly identifies components and an experiment protocol. A broad filing now risks insufficiency and obviousness. Preserve drawings and laboratory decisions from the first engineering iteration so the eventual filing captures the real differentiator.

SECTION 7

Candidate C - novelty-proportional reconstruction and lineage storage

A possible computer-internal improvement, but broad event sourcing and provenance are well known.

7.1 Proposed inventive concept

A storage and reconstruction architecture that stores a seed, versioned finite grammar, exogenous novelty events, and necessary branch reconciliation data instead of storing every intermediate state or projection; it reconstructs state at a requested time, verifies predicate and invariant versions, and uses content-addressed lineage records to preserve splits and merges.

Source note: Source report pp. 23-25, 34, 38, 43, and 51. The source correctly labels novelty-proportional memory as a bounded hypothesis, not a zero-memory theorem.

7.2 Prior-art pressure

The broad ideas are established. Event sourcing stores state changes as events and reconstructs state [IS1]. W3C PROV defines provenance structures [IS2]. Git is a content-addressable store and commits can identify multiple parents [IS3-IS4]. Materialized views persist query results that can be refreshed [IS5]. Many simulation and database systems already use logs, snapshots, checkpoints, deterministic replay, and branch histories.

7.3 Where a technical contribution might remain

- A specific checkpointing or compaction policy based on certified replay cost and branch complexity.
- A data structure that couples event certificates, predicate versions, and lineage hashes to permit deterministic reconstruction with bounded input/output.
- A scheduling method that chooses between recomputation and materialization based on measured expression growth and storage cost.
- A technical application where state reconstruction directly controls or verifies a physical system.

7.4 Patentability assessment

Issue	Assessment
Technical character	Medium if the claims define a concrete memory, storage, input/output, or processor effect; low if they merely define provenance semantics.
Novelty	Low for event log, hashes, parent links, and materialized views.
Inventive step	Low to medium only for a specific reconstruction/checkpoint algorithm with comparative resource data.
Sufficiency	Medium for a restricted deterministic model; low for universal memory claims.
Current readiness	2/5. The source provides schemas and a hypothesis but not a distinct measured storage algorithm.

RECOMMENDED TREATMENT

Initially include the strongest lineage, versioning, and event-certificate features as dependent claims in Candidate A. Split Candidate C into a separate application only when a distinct storage or replay benchmark shows a technical advantage.

SECTION 8

Candidate D - KC projection and prepress pipeline

A narrow image-processing case may exist, but the component techniques are old and heavily populated.

8.1 Proposed inventive concept

A projection or prepress method that maps an output domain into local log-polar coordinates, evaluates an implicit boundary and its gradient, accumulates directional phase-amplitude contributions near the boundary, applies an anisotropic binary threshold or pulse-density policy adapted to log radius and angle, and compensates a calibrated output device or print process to preserve edge placement.

Source note: Source report pp. 30-32, 44, and 50. The report explicitly renames the source's "Feynman vectors" as a phase-amplitude edge accumulator because no path-integral derivation is supplied.

8.2 Prior-art pressure

Logarithmic or log-polar mappings have a long history, including Schwartz's 1977 work on sensory projection [G1]. Signed-distance-field texture rendering was established by Green in 2007 [G2]. Blue-noise halftoning and masks were established by Ulichney and Mitsa/Parker [G3-G4]. Pulse-density modulation for displays and LEDs is known [G5-G6], as is dot-gain compensation in stochastic screening [G7].

The source therefore does not support broad claims to log-polar addressing, SDF edges, 1-bit modulation, blue noise, or dot-gain compensation. A patentable residue would have to be a precise interaction among these elements that produces an objective output improvement.

8.3 Data needed

- Exact accumulator equation, sampling neighborhood, weights, phase definition, and treatment of corners, thin features, and multiple contours.
- Exact anisotropic threshold-mask generation in log-radius and angle, including singularity handling and resampling to the physical Cartesian device.
- Calibration model for display transfer or print dot gain and the sequence by which it modifies the binary decision.
- Comparative metrics: edge displacement, modulation transfer, alias energy, tone error, flicker, print grain, dot-gain error, processing time, and memory.
- A baseline against standard SDF rendering, blue-noise masks, error diffusion, and conventional prepress screening.

8.4 Patentability assessment

Issue	Assessment
Technical character	Medium for concrete display, image-processing, or print-control claims.
Novelty	Low for individual elements; unknown for a precisely defined combination.
Inventive step	Low unless the interaction solves a documented technical problem with a surprising or substantial result.
Sufficiency	Low to medium. The reviewed source gives a pipeline and some equations but not enough implementation detail for broad protection.
Current readiness	1-2/5.

Recommended treatment. Do not make this the first filing. Develop it as a separate image-processing project and file only if a reproducible algorithm and objective improvement can be demonstrated.

SECTION 9

Low-value or non-patent-leading subject matter

Some material may remain useful in a specification, but it should not carry the independent claims.

Subject	Why it should not lead	Better treatment
"Double vacuum" as a name	A label for co-location without coupling. The underlying compatibility mechanism may be claimed, but the metaphor itself provides no patentable technical feature.	Define concrete mode, channel, namespace, policy, or physical isolation conditions.
Quad hourglass	A routing visualization or transition table. Visual organization alone is weak and may be aesthetic or abstract.	Use as a figure or dependent routing embodiment only.
One-bit world model	The source itself narrows one bit to parity/routing/validity. A full world model cannot credibly be reduced to one bit.	Claim the bit's precise control function, not the universe.
Klein bottle / Mobius topology	Known mathematical topology. The source says executable use requires explicit ports, orientation map, quotient rule, and transfer function.	Only claim a concrete device or data-routing implementation with technical effect.
Golden ratio / phi schedule	The source rejects the golden ratio as a correctness requirement and notes symbol overload.	Do not rely on it for novelty or inventive step.
Universal O(1), zero memory/heat/latency	Expressly rejected and physically/computationally implausible at system level.	Remove from patent and investor-facing claims.
Biological, medical, personal, political, or "untouchable identity" claims	The source excludes these as proof or mechanism. They are unsupported and create legal, ethical, and credibility risks.	Exclude entirely unless a separate, enabling, evidence-based invention exists.
The report text, diagrams, code, and names	Protected primarily by copyright, database rights where applicable, confidentiality, and possibly trademarks, not by a patent on the text as a whole.	Use copyright notices, repository records, licenses, and trademark strategy.

SECTION 10

Why the corpus should not be filed as one invention

A shared vocabulary is not the same as one general inventive concept.

10.1 Unity analysis

EPC Article 82 requires one invention or a group linked so as to form a single general inventive concept. At search stage, the examiner asks whether the claims share special technical features that define a contribution over the prior art. The corpus's broad themes - topology, phase, identity, query-first state, projections, and hardware - are too abstract to serve as that feature across all domains.

Pairing	Unity outlook
Software event engine vs. storage	Potentially linked if lineage storage is expressly part of the event certification mechanism; otherwise storage may be a separate invention.
Software event engine vs. Hollowlens-0	Potentially linked if the same support/compatibility/guard protocol is implemented in a specific photonic apparatus. Hardware-specific features may still create a separate invention.
Software event engine vs. KC graphics	Weak link. A downstream projection adapter does not normally share the event solver's special technical feature.
Hollowlens-0 vs. KC graphics/prepress	Different physical systems, problems, and prior art. Almost certainly separate.
All domains plus identity/biology narratives	No credible single general inventive concept.

10.2 Possible umbrella strategy

A carefully drafted first application could describe a common "bounded compatibility event" protocol and include both a software embodiment and one optofluidic embodiment. The independent claims would still need their own technical features and support. If the EPO raises lack of unity, the applicant can pursue divisional applications, but this increases cost and cannot cure missing disclosure. It is safer to draft the initial specification with clearly separable invention modules and abundant fallback positions.

10.3 Why a large omnibus filing can be harmful

- It invites lack-of-unity objections and additional search fees.
- It dilutes the technical problem and makes the independent claims look abstract or aspirational.
- It can disclose multiple valuable projects before each is ready, starting the publication clock and giving competitors a roadmap.
- It may lock in an underdeveloped description; later technical details cannot be added to the original disclosure while retaining priority.
- It makes inventorship and ownership harder because different people may have contributed to different claim groups.

SECTION 11

Representative prior-art landscape

The search shows crowded foundations, not a final anticipation opinion.

11.1 Software and event simulation

Ref.	Reference	Relevant teaching	Impact
PA1	Alur et al., The Algorithmic Analysis of Hybrid Systems (1995)	Continuous variables, discrete modes, guards/transitions; foundational hybrid-systems framework.	Broad state/guard/transition claims are crowded.
PA2	Henzinger, The Theory of Hybrid Automata (1996)	Formal mixed discrete-continuous state systems and algorithmic analysis.	"Phase sheets" or modes do not by themselves distinguish the work.
PA3	Kofman & Junco, Quantized-State Systems (2001)	Discrete-event simulation of continuous-time systems.	Event-driven continuous simulation is not new.
PA4	Redon, Kheddar & Coquillart, Fast Continuous Collision Detection (2002)	Continuous collision detection and event/contact time computation.	Root or time-of-impact solving instead of frame stepping is known.
PP1	US11054807B2	Hybrid automata for cyber-physical systems with continuous/discrete behavior.	Technical applications of hybrid automata are patented.
PP2	US7099809B2	Simulation with state triggers, continuous or user-stepped execution, and post-process state changes.	Triggered simulation architectures are established.
PP3	US8655636B2	Translation of time-driven Simulink diagrams into extended finite automata.	Typed state and event-driven hybrid models are populated.

11.2 Event sourcing, provenance, and materialized state

Ref.	Reference	Relevant teaching	Impact
IS1	Martin Fowler, Event Sourcing (2005)	Every state change captured and stored as an event.	Broad event-log reconstruction is old.
IS2	W3C PROV-DM	Provenance model for entities, activities, and agents.	Generic lineage/provenance semantics are standardized.
IS3-IS4	Git objects and commit-tree documentation	Content-addressable objects; commits with zero, one, or multiple parents.	Hashes, branches, and parent lineage are not new.
IS5	PostgreSQL materialized views	Persisted query results that can be refreshed.	Projection-as-materialized-view is known terminology and architecture.

11.3 Optofluidic and programmable photonic systems

Ref.	Reference	Relevant teaching	Impact
PH1	US10875020B2	Reconfigurable microvalve optical waveguide and liquid tuning.	Liquid-tuned waveguide components are known.
PH2	US12044623B2	Optofluidic analyte detection using MMI waveguides.	Optofluidic detection through waveguides is known.
PH3	US10359272B2	Programmable photonic processing and linear optical transformations.	Programmable photonic transfer matrices are crowded.
PH4	US20170031101A1	Automatic adjustment of optical interferometers for improved extinction ratio.	Self-calibration and control of interferometers are known.

Ref.	Reference	Relevant teaching	Impact
PH5	WO2022168074A1	Calibration, monitoring, and control of integrated photonic systems.	A generic digital calibration sidecar is not enough.

11.4 Graphics, halftoning, and display control

Ref.	Reference	Relevant teaching	Impact
G1	Schwartz, Spatial Mapping in the Primate Sensory Projection (1977)	Logarithmic conformal/log-polar mapping.	Log-polar mapping is old.
G2	Green, Improved Alpha-Tested Magnification for Vector Textures (2007)	Signed-distance fields stored in textures for edge reconstruction.	SDF edge rendering is old.
G3	Ulichney, Dithering with Blue Noise (1988)	Blue-noise dithering for binary displays.	Blue-noise output is old.
G4	Mitsa & Parker, Digital Halftoning Using a Blue-Noise Mask (1992)	Pixelwise thresholding with blue-noise masks.	Blue-noise masks are old.
G5	US11640805B2	Delta-sigma pulse-density modulation for digital displays.	PDM display control is known.
G6	US20150245438A1	PDM of LED current for dimming and spectrum shaping.	PDM LED drive is known.
G7	WO2009023086A2	Stochastic halftone screening and dot-gain compensation.	Dot-gain compensation is known.

SEARCH CONCLUSION

The prior art does not prove that every narrow implementation is unpatentable. It does show that the source's building blocks are largely known. Any viable claim must identify a specific interaction, technical effect, and implementation detail not taught or suggested by the combined landscape.

SECTION 12

Inventorship, ownership, and AI-assisted drafting

A name on a corpus is not automatically the inventor of every claim.

12.1 Claim-specific inventorship

The report says it was prepared from seven user-supplied PDFs and also identifies report-level normalizations, such as event sourcing, materialized views, provenance graphs, and query planning, that were not consistently used in the sources. This matters. A future claim may combine source-originated concepts, later human corrections, and report-level drafting. Inventorship must be traced to the human who conceived the claimed inventive features, not merely to the person who assembled or owns the documents.

12.2 Required contribution audit

- For every proposed independent claim, list each substantive limitation and the human contributor who conceived it.
- Record the date and evidence: notebook page, source PDF page, commit, email, drawing, prototype, or meeting note.
- Distinguish conception from routine implementation, testing, funding, editing, and project management.
- Identify employment, contractor, university, or collaboration agreements that may assign rights.
- Execute assignments before filing where the applicant differs from the inventor.

12.3 AI assistance

Under current EPO law, a machine cannot be designated as inventor [E12]. AI can assist with analysis, drafting, searching, coding, and visualization, but human inventorship still depends on human contribution to the inventive concept. Preserve a record of what existed before AI assistance and which technical choices were made by humans. Do not assume an AI-generated combination is automatically owned, novel, or sufficiently disclosed.

PRACTICAL RULE

Before sending claim drafts to counsel, complete the inventorship matrix in Appendix B. Do not list or omit inventors for convenience; incorrect inventorship can create entitlement and enforceability disputes.

SECTION 13

Disclosure and novelty risk

This is the most time-sensitive issue because a lost novelty position cannot usually be repaired.

13.1 Immediate concern

The source report is dated 14 August 2026 and refers to seven earlier PDFs, a prototype, diagrams, source maps, and a ZIP package. The reviewed PDF does not establish whether any of these materials have been posted online, emailed without confidentiality, presented, sold, demonstrated, deposited in a public repository, or otherwise made available to the public. Each event must be audited separately.

13.2 AI-platform issue

Octrooiencentrum Nederland specifically lists entering sensitive information into AI as an example of disclosure to avoid before filing [NL1]. This assessment cannot determine whether this conversation or any prior AI use is legally public or confidential. Do not assume that using an AI service preserves novelty. Give patent counsel the relevant service terms, account settings, dates, recipients, and the exact material uploaded so counsel can assess the risk.

13.3 Disclosure audit questions

1. What is the earliest date on which each of the seven source PDFs was sent outside a confidentiality obligation?
2. Were any PDFs, excerpts, code, diagrams, demos, videos, repositories, or product descriptions publicly accessible?
3. Were conference, investor, supplier, university, customer, or grant discussions protected by an NDA or another enforceable duty of confidence?
4. Was any device sold, offered for sale, used publicly, or made available for inspection?
5. Which exact technical features were disclosed in each event? A disclosure may destroy some claims but not later, undisclosed improvements.
6. Did any third party contribute or independently publish similar material before the intended filing date?
7. Have prior patent applications been filed by any contributor, employer, or collaborator?

13.4 Response if disclosure occurred

Do not conclude that all rights are lost without legal analysis. Counsel should compare the disclosed subject matter with each proposed claim, determine whether the disclosure was actually public, assess confidentiality, and consider jurisdictions with grace periods. For Europe, however, the ordinary position is strict absolute novelty; the EPC exceptions are narrow [E5-E6]. A later undisclosed improvement may still be protectable if it is independently new and inventive.

IMMEDIATE ACTION

Stop new non-confidential disclosures. Create the disclosure log in Appendix A today. Contact an octrooigemachtigde before publishing code, circulating the report, demonstrating hardware, or sending unrestricted investor materials.

SECTION 14

Filing routes and portfolio sequence

Use priority strategically, but do not file an under-enabled concept simply to obtain a date.

14.1 Suggested sequence

Timing	Stage	Deliverable
Now	Confidential invention harvest	Disclosure audit; inventorship map; split A/B/C/D; preserve dated evidence; stop public release.
Weeks 1-4	Technical evidence sprint	Candidate A benchmark and Candidate B engineering specification; decide whether any disclosure deadline requires an urgent filing.
First filing	Dutch or direct European filing	Use a professionally drafted specification with multiple fallback positions. A Dutch first filing can establish priority and provide a search.
By 12 months	EP/PCT decision	Use search results, prototype data, business case, and disclosure developments to choose European, PCT, or selected national routes [NL5].
After filing	Continue R&D under change control	File later improvement applications for new matter; do not assume later additions receive the original priority date.

14.2 Dutch first filing versus direct EPO

A Dutch first filing may be useful for establishing priority and obtaining a search while preserving options. The Dutch application is published and granted at 18 months, and current Dutch practice grants without substantive examination even if the search is unfavorable; this makes claim quality and later validity analysis especially important [NL3, NL7]. A direct EPO filing provides substantive examination but normally costs more and demands a stronger initial specification. Counsel should choose based on disclosure urgency, target markets, budget, and whether the invention is mature enough for European scrutiny.

14.3 PCT

A PCT application does not itself grant a worldwide patent. It can defer national/regional choices and costs, generally while preserving the first filing's priority if filed within 12 months. It is useful only when the business case justifies multi-country protection and the initial disclosure is strong enough to support later claims.

14.4 Claim-category planning

For Candidate A, the EPO permits aligned independent claims in method, apparatus/system, computer program, and computer-readable-medium form when properly drafted [E13]. Candidate B can include apparatus, operating method, calibration method, and controller claims. Candidate D can include image-processing method, device/controller, program, medium, and possibly a manufactured print or signal only where the technical relationship is adequately supported.

SECTION 15

Freedom to operate and complementary protection

A patent is a right to exclude, not an automatic right to practise.

15.1 Patentability is not freedom to operate

Even a patentable improvement can fall within broader third-party patents. Candidate B is particularly exposed because liquid-tuned waveguides, interferometers, programmable photonic meshes, detection, and calibration are active patent areas. Candidate D may encounter display, halftoning, print-screening, and device-control patents. A separate FTO search should be based on the final product architecture, target countries, and live claims, not merely on similar titles or expired academic publications.

15.2 Complementary IP

Tool	Best use here	Caution
Copyright	Source texts, diagrams, code, schemas, and documentation	Automatic protection of expression; does not protect the underlying technical idea.
Trade secret	Solver heuristics, compatibility policies, calibration coefficients, datasets, manufacturing tolerances	Useful where secrecy can be maintained and reverse engineering is difficult.
Trademark	Names such as Equation World Zero, Hollowlens-0, KC Vector, or product branding	Protects source identifiers, not technical functionality. Search and clearance required.
Design right	Product or interface appearance	May complement hardware/display products; not a substitute for functional protection.
Contract	NDAs, assignments, collaborator terms, source-code licences	Essential for preserving confidentiality and ownership.
Publication	Defensive publication of non-core variants	Can block later patents by others but also destroys the publisher's own novelty; use only strategically after filing decisions.

SECTION 16

Thirty-day action plan

Convert the corpus into an attorney-ready invention package before more disclosure.

Days 1-3: preserve rights

- Freeze public uploads, public repositories, unrestricted distribution, demonstrations, and marketing claims.
- Complete the disclosure audit for every source PDF, codebase, image, demo, and conversation.
- Create a secure, access-controlled evidence folder with original timestamps, hashes, and contributor records.
- List all human contributors and contractual relationships.

Days 4-10: split inventions

- Write a two-page invention disclosure for Candidate A using the checklist below.
- Write a separate physical specification for Candidate B, including annotated system drawings and a parameter table.
- Decide whether Candidate C is merely support for A or a distinct storage invention.
- Place Candidate D on hold unless a concrete algorithm and test images already exist.

Days 8-21: evidence sprint

- Implement Candidate A in at least two relation families and one technical application.
- Run equal-correctness benchmarks against frame stepping and a conventional event baseline.
- Record failure cases: tangencies, coincident roots, branch explosion, compatibility changes, and uncertainty intervals.
- For Candidate B, complete optical/fluid/controller design calculations and, if feasible, a benchtop measurement or simulation tied to a fabrication plan.

Days 15-30: counsel and filing decision

- Engage a Dutch/European patent attorney experienced in computer-implemented inventions and photonics/optofluidics.
- Request a confidential invention-harvesting session and a targeted search for Candidate A and Candidate B.
- Ask Octrooiencentrum Nederland about confidential advice and the free orienting search where eligible [NL4].
- Choose first filing route and applicant; finalize assignments and inventor declarations.
- Prepare a public-communications embargo plan through the filing date.

Candidate A minimum evidence pack

- Define the technical problem, typed physical state, exact support/compatibility/root/transition algorithm, and correctness boundary.
- Provide at least two equal-correctness baselines and record root calls, latency distribution, memory or IO, event errors, and kill criteria.
- Document alternative relation families, support representations, compatibility schemas, solver implementations, and technical-system embodiments.

SECTION 17

Claim-drafting skeletons

These are invention-harvesting prompts, not filing-ready claims.

DO NOT FILE THESE VERBATIM

The skeletons deliberately expose the technical features that need evidence. A patent attorney must rewrite them after a full search, inventorship review, and jurisdiction-specific analysis.

17.1 Candidate A - method skeleton

1. receiving a typed state representation of a physical system, the state including continuous physical variables and a discrete operating-mode identifier;
2. obtaining a plurality of relation functions defining respective candidate transition surfaces and respective analytic support functions;
3. for each candidate transition, evaluating the support function and a versioned compatibility predicate to reject a candidate before numerical root refinement when the state is outside the support or the operating modes are incompatible;
4. for each non-rejected candidate, applying a certified root procedure over a bounded time interval to determine a root or root interval of the relation function;
5. selecting an earliest admitted root according to a deterministic ordering rule;
6. generating an event certificate containing at least the relation identifier, predicate version, root interval or confidence measure, and a pre-transition state digest;
7. applying a typed state transition, verifying a declared invariant, generating a post-transition state digest, and storing a lineage relation;
8. wherein the support and compatibility evaluations reduce a measured numerical-solver workload at a stated event-detection error boundary relative to a baseline method.

17.2 Candidate A - dependent-feature menu

- Support is an annular sector, cone, shell, causal region, collision region, or device-specific operating envelope.
- Compatibility uses sheet/mode, phase interval, address, policy, channel, time window, or invariant tag and returns reason codes.
- Root certification uses interval arithmetic, analytic roots, polynomial isolation, conservative advancement, or a verified numerical bound.
- The event certificate includes multiplicity, tangency flag, uncertainty, branch creation, parity before/after, and invariant report.
- The physical system is a robot, machine tool, vehicle, optical circuit, electrical network, process plant, or other defined technical system.
- A scheduler orders support/compatibility checks using cached predicates or estimated solve cost.

17.3 Candidate B - apparatus skeleton

1. an optical source and input interface configured to illuminate a defined angular or spatial support;
2. a fluidically tunable optical coupling region having specified optical and fluidic parameters;
3. an interferometric waveguide circuit having at least two output paths and balanced photodetection;
4. an actuator and calibration subsystem configured to adjust a coupling or phase parameter and to estimate drift;
5. a controller storing a channel-compatibility rule, a measured guard condition, and a confidence threshold;

6. the controller declaring an event only when the input is within the support, the selected channel is compatible, the balanced detector signal crosses the guard, and the confidence threshold is met;
7. the controller storing a calibrated event record and suppressing or flagging an event when a reference-channel or drift condition fails;
8. wherein the arrangement provides a measured reduction in crosstalk, false events, calibration energy, or energy per verified event relative to a matched baseline.

17.4 Candidate C - storage skeleton

1. storing a seed, a versioned deterministic grammar, and an append-only stream of exogenous events for a technical-system model;
2. storing branch records with parent digests, reconciliation rules, and predicate/invariant versions;
3. reconstructing state at a requested time by evaluating a closed expression and replaying only admitted exogenous events and necessary branch records;
4. selectively materializing a checkpoint when an estimated replay cost exceeds a storage or latency threshold;
5. verifying reconstructed state against content digests and event certificates;
6. wherein the selection reduces storage input/output or reconstruction latency relative to periodic full-state snapshots at an equivalent correctness boundary.

17.5 Candidate D - image-processing skeleton

1. mapping image or print coordinates into a local log-polar coordinate domain;
2. evaluating a signed-distance or implicit boundary function and a local gradient;
3. accumulating weighted directional contributions in a neighborhood of the boundary according to a defined phase-amplitude rule;
4. selecting a binary output using an anisotropic threshold structure parameterized by log radius, angle, and boundary gradient;
5. applying a calibrated device or print-process compensation function;
6. mapping the binary result to a physical display or print grid while controlling singularity and resampling error;
7. wherein measured boundary displacement, alias energy, flicker, tone error, or dot-gain error is reduced relative to a stated baseline.

SECTION 18

Source-to-claim map and references

Traceability for counsel and for the next technical drafting round.

18.1 Source-to-candidate map

Family	Source pages	Content
Candidate A	pp. 4-5	Mature event-calculus synthesis and bounded feasibility verdict
Candidate A	pp. 14-18	Typed state, event equation, event record, compatibility / double vacuum
Candidate A	pp. 25-27	Novelty log, query API, execution plan, solver obligations
Candidate A	pp. 34-38	Equation World Zero, worked trace, benchmark matrix
Candidate A	pp. 41-47	Risks, claim matrix, recommended build and exit criterion
Candidate B	pp. 39-40	Hollowlens-0 architecture and matched-baseline requirement
Candidate B	pp. 45 and 50-53	Hardware translations, liquid substrate, control sidecar, metric
Candidate C	pp. 23-25 and 51	Information-science normal form, lineage, log, novelty memory
Candidate D	pp. 29-32 and 50	Graphics boundary, KC adapter, log-polar/SDF/quantization pipeline
Non-candidates	pp. 10-11, 19-21, 44-45, 48	Demoted topology and rejected totalizing/medical claims

18.2 Legal and official references

- [E1] EPC Article 52 - Patentable inventions. <https://www.epo.org/en/legal/epc/2020/a52.html> Accessed 14 August 2026.
- [E2] EPC Article 54 - Novelty. <https://www.epo.org/en/legal/epc/2020/a54.html> Accessed 14 August 2026.
- [E3] EPC Article 56 - Inventive step. <https://www.epo.org/en/legal/epc/2020/a56.html> Accessed 14 August 2026.
- [E4] EPC Article 83 - Disclosure of the invention. <https://www.epo.org/en/legal/epc/2020/a83.html> Accessed 14 August 2026.
- [E5] EPC Article 82 - Unity of invention. <https://www.epo.org/en/legal/epc/2020/a82.html> Accessed 14 August 2026.
- [E6] EPC Article 55 - Non-prejudicial disclosures. <https://www.epo.org/en/legal/epc/2020/a55.html> Accessed 14 August 2026.
- [E7] EPO Guidelines 2026, G-II 3.3 - Mathematical methods. https://www.epo.org/en/legal/guidelines-epc/2026/g_ii_3_3.html Technical contribution through application or specific implementation.
- [E8] EPO Guidelines 2026, G-II 3.3.2 - Simulation, design or modelling. https://www.epo.org/en/legal/guidelines-epc/2026/g_ii_3_3_2.html Simulation assessed under ordinary technical-effect principles.
- [E9] EPO Guidelines 2026, G-II 3.6.1 - Further technical effects. https://www.epo.org/en/legal/guidelines-epc/2026/g_ii_3_6_1.html Examples include control and computer-internal operation.
- [E10] EPO Guidelines 2026, G-VII 5.4 - Mixed technical/non-technical claims. https://www.epo.org/en/legal/guidelines-epc/2026/g_vii_5_4.html COMVIK approach.
- [E11] EPC Article 60 - Right to a European patent. <https://www.epo.org/en/legal/epc/2020/a60.html> Inventor or successor in title; employee rules.
- [E12] EPO J 0008/20 - Designation of inventor / DABUS. <https://www.epo.org/en/boards-of-appeal/decisions/j200008eu1> A machine is not an inventor under the EPC.
- [E13] EPO Guidelines 2026, F-IV 3.9.1 - CII claim formulations. https://www.epo.org/en/legal/guidelines-epc/2026/f_iv_3_9_1.html Method, system, program, and medium formulations.
- [NL1] Octrooiencentrum Nederland - Voorwaarden voor octrooi. <https://www.octrooiencentrum.nl/octrooien/orienteren-op-octrooi/voorwaarden-voor-een-octrooi> Novelty and secrecy; includes AI disclosure warning.
- [NL2] Octrooiencentrum Nederland - Nederlands octrooi: acties en valkuilen. <https://www.octrooiencentrum.nl/octrooien/orienteren-op-octrooi/nederlands-octrooi-acties-en-valkuilen> Disclosure, search, weak applications, priority year.

- [NL3] Octrooicentrum Nederland - Nederlands octrooi aanvragen. <https://www.octrooicentrum.nl/octrooien/octrooi-aanvragen-en-verlening/nederlands-octrooi-aanvragen> Search, publication, grant, duration.
- [NL4] Octrooicentrum Nederland - Oriënterend octrooionderzoek. <https://www.octrooicentrum.nl/voorlichting/oriënterend-octrooionderzoek> Confidential orienting search information.
- [NL5] Octrooicentrum Nederland - Octrooi aanvragen in het buitenland. <https://www.octrooicentrum.nl/octrooien/octrooi-aanvragen-en-verlening/octrooi-aanvragen-het-buitenland-met-nederlands-octrooi> European/PCT routes and 12-month priority year.
- [NL7] Octrooicentrum Nederland - Landen kiezen voor je octrooi. <https://www.octrooicentrum.nl/octrooien/oriënteren-op-octrooi/landen-kiezen-voor-je-octrooi> Dutch unexamined grant and European examined route.

18.3 Representative prior-art references

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- [PA4] S. Redon, A. Kheddar & S. Coquillart, Fast Continuous Collision Detection between Rigid Bodies (2002). <https://hal.inria.fr/inria-00390356/fr/>
- [PP1] US11054807B2 - Hybrid automata mining from cyber-physical traces. <https://patents.google.com/patent/US11054807B2/en>
- [PP2] US7099809B2 - Modeling system. <https://patents.google.com/patent/US7099809B2/en>
- [PP3] US8655636B2 - Semantic translation of time-driven Simulink diagrams. <https://patents.google.com/patent/US8655636B2/en>
- [IS1] Martin Fowler, Event Sourcing. <https://martinfowler.com/eaDev/EventSourcing.html>
- [IS2] W3C PROV-DM: The PROV Data Model. <https://www.w3.org/TR/prov-dm/>
- [IS3] Git Book - Git Objects. <https://git-scm.com/book/en/v2/Git-Internals-Git-Objects>
- [IS4] Git documentation - git-commit-tree. <https://git-scm.com/docs/git-commit-tree>
- [IS5] PostgreSQL documentation - Materialized Views. <https://www.postgresql.org/docs/current/rules-materializedviews.html>
- [PH1] US10875020B2 - Reconfigurable microvalve optical waveguide. <https://patents.google.com/patent/US10875020B2/en>
- [PH2] US12044623B2 - Optofluidic analyte detection using MMI waveguides. <https://patents.google.com/patent/US12044623B2/en>
- [PH3] US10359272B2 - Programmable photonic processing. <https://patents.google.com/patent/US10359272B2/en>
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- [G5] US11640805B2 - Delta-sigma PDM for a digital display. <https://patents.google.com/patent/US11640805B2/en>
- [G6] US20150245438A1 - PDM modulation of LED current. <https://patents.google.com/patent/US20150245438A1/en>
- [G7] WO2009023086A2 - Stochastic halftone images based on screening parameters. <https://patents.google.com/patent/WO2009023086A2/en>

SECTION A-D

Practical templates and checklists

Use these to prepare a confidential invention package for counsel.

Appendix A - disclosure audit log

Date	Material/feature	Recipient/location	Public or confidential?	NDA / duty	Evidence	Counsel assessment

Complete one line for every website, repository, email, meeting, pitch, conference, sale, demonstration, AI upload, supplier discussion, and source-document circulation. Preserve screenshots, access controls, recipient lists, NDAs, and dates.

Appendix B - inventorship contribution matrix

Claim feature	Human contributor	Date conceived	Evidence	Employer/contract	Notes

Appendix C - experiment record checklist

- ☐ Repository commit or immutable archive identifier
- ☐ Exact hardware/software environment
- ☐ Input datasets or model parameters and provenance
- ☐ Baseline implementations and parameter parity
- ☐ Correctness/error definition and uncertainty bounds
- ☐ Candidate counts, solver calls, latency distribution, memory, IO, and energy where applicable
- ☐ Failure cases and kill criteria
- ☐ Raw results, scripts, plots, and negative results
- ☐ Human contributors and dates
- ☐ Confidentiality classification and intended filing family

Appendix D - questions for patent counsel

1. Based on the disclosure log, which Candidate A/B/C/D features remain novel in Europe and other target jurisdictions?
2. Should the first filing be Dutch, direct EPO, or another route, and what must be in the filing to support the 12-month strategy?
3. Does Candidate A have a credible technical effect under current EPO computer-implemented-invention practice?
4. Can Candidate A and B share one general inventive concept, or should they be filed separately from the outset?
5. Which prior-art classes and competitors should be searched more deeply?
6. Who are the inventors for each independent claim, and what assignments are required?
7. Which features should remain trade secrets rather than be disclosed in a patent?
8. What product architecture should be searched for freedom to operate, and in which countries?
9. Is an urgent filing needed before any planned publication, pitch, demonstration, grant application, or collaboration?
10. What evidence and examples are necessary to avoid an Article 83 sufficiency objection?

END OF PRELIMINARY ASSESSMENT

Prepared 14 August 2026 from the supplied 63-page source report and representative public sources current on that date.